

OFFLINE PREPARATION AND DEPLOYMENT

102-run dataset



run-level train/validation split

Nonlinear AGV model



implemented discrete transition

training/validation truth is not used online

Train · calibrate · linearize



offline identification pipeline

W

Weights + stats

ModernTCN deployment

E

Error moments

P_m, P_o, C by regime

L

LPV database

4125 local models

ONLINE PERCEPTION → CONTROL LOOP

22-D feature bank

ModernTCN-Delta

z_k



$z = \Delta_1 \Delta_2 \Delta_4$

$X_{128 \times 88}$

grade + confidence

θ^M, c^{main}

Onboard sensing



CAN · yaw · six-axis IM

Qualified observer



IMU · innovation qualification

θ^{obs}, q_{obs}

Uncertainty-adaptive fusion



NIS gate · bounded ± 0.5

Grade conditioning



deadband · clip ± 10

$T_s = 0.01$ s · one update per control period



motion command

r^{ref}

Fused-grade LPV-MPC

model

d

F_{eq}

Q, R

bounds

u_k

AGV plant



y_{k+1}

temporal primary

observer auxiliary

offline deployment